ShinyMR Promotes Reproducible and Rigorous Mendelian Randomization Studies with GWAS Summary Statistics

AUTHOR

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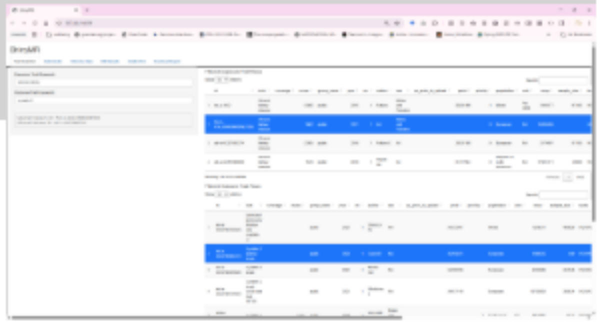
Problem statement

- Two-Sample MR is widely used with GWAS summary statistics like those in the OpenGWAS database and the GWAS Catalog (Davey Smith & Ebrahim, 2003; Hartwig et al., 2016).
- Ease of data access has fueled proliferation of MR studies that lack rigor and may provide misleading results (Elsworth et al., 2020; Lu et al., 2025; MacArthur et al., 2017).
- We developed ShinyMR (https://github.com/fboehm/shinyMR) to combat this problem.

User interface



ShinyMR provides a reproducible Rmarkdown file for primary MR analysis and sensitivity analyses to guide assessments of biases due to assumption violations

References

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https://github.com/fboehm/shinyMR